Lesson 19 More on Relations

In Lesson 18, we introduced equivalence relations and gave a theorem which led to the important fact that an equivalence relation R on a set A defines a partition on A. We will prove that theorem now.

Theorem 2 (Lesson 18)
Let R be an equivalence relation on a set A, and let $a, b \in A$.

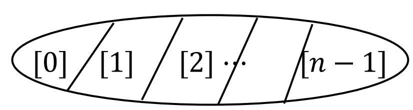
a) $a \in [a]$
b) aRb if and only if $[a] = [b]$
c) If $[a]$ and $[b]$ are not disjoint, then $[a] = [b]$.

Proof

a) Since R is an equivalence relation, it is reflexive, so aRa, and $a \in [a]$.
b) ($\Longrightarrow$) Suppose aRb, we show that $[a] = [b]$, i.e. $[a] \subseteq [b]$ and $[b] \subseteq [a]$.
Let $x \in [a]$, then xRa. Since R is transitive (R is an equivalence relation), xRa and aRb imply xRb, and $x \in [b]$. Therefore, $[a] \subseteq [b]$.
Let $x \in [b]$, then xRb. As R is symmetric (R is an equivalence relation), aRb implies bRa. Since R is transitive, xRb and bRa imply xRa, so $x \in [a]$ and $[b] \subseteq [a]$.
($\Longleftarrow$) From (a), we have $a \in [a]$. If $[a] = [b]$, then $a \in [b]$, and aRb.
c) If $[a] \cap [b] \neq \emptyset$, then $[a] = [b]$.
If $[a] \cap [b] \neq \emptyset$, then there is an element $x \in [a] \cap [b]$. So, $x \in [a]$ and $x \in [b]$, i.e. xRa and xRb. Since R is symmetric, xRa implies aRx. Since R is transitive, aRx and xRb imply aRb, and by (b), we have $[a] = [b]$.

In Lesson 18, we saw that the "congruence modulo 5" relation on $\mathbb{Z}$ partitions $\mathbb{Z}$ into 5 equivalence classes. In general, for the "congruence modulo n" relation on $\mathbb{Z}$, there are n equivalence classes:

$[a] = \{kn + a : k \in \mathbb{Z}\}, \quad a = 0, 1, 2, \dots, n-1$



The classes are called ***congruence classes modulo n*** and the set of congruence classes modulo n is denoted $\mathbb{Z}_n$:

$$\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}$$

We can prove that if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a \pm c \equiv b \pm d \pmod{n}$ and $ac \equiv bd \pmod{n}$ (Exercises). By Theorem 2(b) in Lesson 18, if $[a] = [b]$ and $[c] = [d]$, then $[a \pm c] = [b \pm d]$ and $[ac] = [bd]$, i.e. addition, subtraction, and multiplication are well-defined on $\mathbb{Z}_n$. We may treat $\mathbb{Z}_n$ as a number system with n "numbers": $[0], [1], [2], \dots, [n-1]$ and do **modular arithmetic** on $\mathbb{Z}_n$. For example, when $n = 5$, $[2] + [4] = [6] = [1]$, $[2][4] = [8] = [3]$ etc. Division is trickier, we discuss it in an optional exercise.

Finally, we discuss two new relations, they are ordering relations, generalize the "less than or equal to" ($\leq$) relation on numbers.

Definition A ***partial order*** R on a set A is a relation on A such that:

- xRx for all $x \in A$ (i.e. R is reflexive)
- if xRy and yRz, then xRz, for all $x, y, z \in A$ (i.e. R is transitive)
- if xRy and yRx, then $x = y$, for all $x, y \in A$ (this property is ***anti-symmetry***)

A partial order is called a ***total order*** if, in addition,

- xRy or yRx for all $x, y \in A$

In general, a partial order is NOT an equivalence relation, as it is not required to be symmetric.

a) The "$=$" relation on $\mathbb{Z}$ is reflexive and transitive (Example 7a in Lesson 17). Moreover, for all $x, y \in \mathbb{Z}$, if $x = y$ and $y = x$, then $x = y$, so "$=$" is a partial order. It is not a total order: $1 \neq 2$ and $2 \neq 1$.
b) The "$\leq$" relation on $\mathbb{Z}$ is reflexive and transitive (Example 7b in Lesson 17). For all $x, y \in \mathbb{Z}$, if $x \leq y$ and $y \leq x$, then $x = y$, so is a partial order. It is a total order as $x \leq y$ or $y \leq x$ for all $x, y \in \mathbb{Z}$.
c) Let A be a set. The subset relation on $\mathcal{P}(A)$ is reflexive and transitive (Example 7d in Lesson 17). For all $S, T \in \mathcal{P}(A)$, if $S \subseteq T$ and $T \subseteq S$, then $S = T$, so $\subseteq$ is a partial order. It is not a total order because it is not true that we must have $S \subseteq T$ or $T \subseteq S$. For example, $S = \{1\}$ and $T = \{2\}$ are subsets of $A = \{1,2\}$, but $S \nsubseteq T$ and $T \nsubseteq S$.